

# Homework 6

Alejandro De La Torre  
MATH 421 — The Theory of Single Variable Calculus  
Section 003  
Instructor: Grace Work

October 2025

## Collaboration Statement

---

**Problem 1** ( Ross Chapter 2, 9.3 )

---

Suppose  $\lim a_n = a$ ,  $\lim b_n = b$ , and  $s_n = \frac{a_n^3 + 4a_n}{b_n^2 + 1}$ . Prove that  $\lim s_n = \frac{a^3 + 4a}{b^2 + 1}$  carefully, using the limit theorems.

### Discussion.

Here we essentially want to break the problem up into the individual limit components and apply the limit theorems 9.1-9.7 accordingly to show that the limit converges to  $\frac{a^3 + 4a}{b^2 + 1}$ .

For reference, here is what each of the 9.1-9.7 theorems say in a nutshell:

9.1 Theorem: Convergent sequences are bounded.

9.2 Theorem: If  $\lim s_n = s$ , then  $\lim(ks_n) = k \lim s_n$ .

9.3 Theorem: If  $\lim s_n = s$  and  $\lim t_n = t$ , then  $\lim(s_n + t_n) = \lim s_n + \lim t_n$ .

9.4 Theorem: If  $\lim s_n = s$  and  $\lim t_n = t$ , then  $\lim(s_n t_n) = (\lim s_n)(\lim t_n)$ .

9.5 Theorem: If  $\lim s_n = s$ , ( $s \neq 0$ ) and  $s_n \neq 0$  for all  $n$ , then  $\lim \frac{1}{s_n} = \frac{1}{s}$ .

9.6 Theorem: If  $\lim s_n = s$ , ( $s \neq 0$ ),  $s_n \neq 0$  for all  $n$ , and  $\lim t_n = t$ , then  $\lim \frac{t_n}{s_n} = \frac{t}{s}$ .

9.7 Theorem: (a)  $\lim \frac{1}{n^p} = 0$  for  $p > 0$ .

(b)  $\lim a^n = 0$  if  $|a| < 1$ .

(c)  $\lim(n^{\frac{1}{n}}) = 1$ .

(d)  $\lim(a^{\frac{1}{n}}) = 1$  for  $a > 0$ .

### Proof.

Given that  $\lim a_n = a$  and  $\lim b_n = b$ , we can then express  $\lim \frac{a_n^3 + 4a_n}{b_n^2 + 1}$  as  $\frac{\lim a_n^3 + \lim 4a_n}{\lim b_n^2 + \lim 1}$ . We can

express the result of  $\lim a_n^3$  as  $a^3$ , and  $\lim b_n^2 = b^2$  by theorem 9.4. Next we can show that  $\lim 4a_n = 4a$  by theorem 9.2. Then, using the result from theorem 9.3, we can show that  $\lim(a_n^3 + 4a_n) = a^3 + 4a$  and  $\lim(b_n^2 + 1) = b^2 + 1$ . Finally, using the result from theorem 9.6, we show that:

$$\lim s_n = \frac{\lim(a_n^3 + 4a_n)}{\lim(b_n^2 + 1)} = \frac{a^3 + 4a}{b^2 + 1}$$

for  $b^2 + 1 \neq 0$  for all  $n$ . □

**Problem 2** (*Ross Chapter 2, 9.9(b)*)

Suppose there exists  $N_0$  such that  $s_n \leq t_n$  for all  $n > N_0$ . Prove that if  $\lim t_n = -\infty$ , then  $\lim s_n = -\infty$ .

**Discussion.**

Here we want to leverage the definition of divergence which is given in Ross 9.8 theorem, which states that:

*We write  $\lim s_n = -\infty$  provided for each  $M < 0$  there is a number such that  $n > N$  implies  $s_n < M$ .*

The nuance here is that we are comparing a sequence diverging to a 'negative infinity' ( $t_n$ ) to another sequence that we are given to suppose is 'more negative' ( $s_n$ ) after a certain point ( $N_0$ ; for all  $n > N_0$ ). In other words, we are comparing a sequence diverging to negative infinity ( $t_n$ ) with another sequence ( $s_n$ ) that is eventually less than or equal to it.

So if  $\lim t_n = -\infty$  is true, then we know by theorem 9.8 that for every  $M < 0$  there is a number  $N_1$  such that  $n > N_1$  implies  $t_n < M$ . We also know that there has got to be a  $N_0$  (we suppose there is) such that  $s_n \leq t_n$  for all  $n > N_0$ , so we can then just define an  $N$  that is greatest of the set of  $\{N_0, N_1\}$  in whatever arbitrary case we take on (which we know much exist) so that we can say  $s_n \leq t_n \leq M$  so that we can henceforth conclude that  $s_n < M$  which is what we need to say that  $s_n$  diverges.

**Proof.**

Let  $M < 0$ . We know that there exists an  $N_1$  such that  $n > N_1$  implies  $t_n \leq M$ .

Let  $N = \max\{N_0, N_1\}$ . Thus,  $n > N$  implies

$$s_n \leq t_n \leq M$$

which implies

$$s_n \leq M$$

Therefore  $\lim s_n = -\infty$  □

**Problem 3** (*Ross Chapter 2, 9.10(b)*)

Show that  $\lim s_n = +\infty$  if and only if  $\lim(-s_n) = -\infty$ .

**Discussion.** We want to utilize theorem 9.8 and use a distinct  $M$  according to each case (since this is an iff statement) and prove divergence accordingly.

We will begin by show it first in the forward (  $\implies$  ) direction: "if  $\lim s_n = +\infty$ , then  $\lim(-s_n) = -\infty$ , wherein we would let  $M < 0$  so that we can say  $s_n > -M$  then manipulate the expression to arrive at the conclusion that if  $\lim s_n = +\infty$  then  $\lim(-s_n) = -\infty$ .

Then we would show the backward (  $\impliedby$  ) direction of the iff statement such that: "if  $\lim(-s_n) = -\infty$  then  $\lim s_n = +\infty$ ", and follow the same procedure but with  $M > 0$  to show the divergence toward a positive  $\infty$  result.

**Proof.**

(  $\implies$  ) Let  $\lim s_n = -\infty$ . Let  $M < 0$ . For each  $M < 0$  there is a number  $N$  such that  $n > N$  implies

$$s_n > -M$$

which implies

$$-s_n < M$$

Thus,  $\lim(-s_n) = -\infty$

(  $\impliedby$  ) Let  $\lim(-s_n) = -\infty$ . Let  $M > 0$ , For each  $M > 0$  there us a number  $N$  such that  $n > N$  implies

$$(-s_n) > -M$$

which implies

$$s_n > M$$

Thus,  $\lim(s_n) = \infty$  □

#### **Problem 4** ( Ross Chapter 2, 10.2 )

*Prove Theorem 10.2 for bounded decreasing sequences.*

**Discussion.**

Theorem 10.2 states that "All bounded monotone sequences converge." In the textbook, this theorem is proven first for the case of a bounded increasing sequence, using the concept of the supremum (*sup*). To prove the theorem for a bounded decreasing sequence, we instead use the *infimum* (*inf*) as our central idea. Intuitively, the argument mirrors the increasing case: rather than the terms of the sequence being squeezed upward toward the supremum, here they are squeezed downward toward the infimum.

So, given a bounded decreasing sequence  $(s_n)$ , we recognize that since it is bounded below, there exists an  $\inf S$ , say  $L$ , which acts as the "greatest lower bound" of the set  $\{s_n : n \in \mathbb{N}\}$ . The proof strategy is to show that  $s_n$  converges to this infimum  $L$  using the  $\varepsilon$ -definition of convergence, just as the increasing case shows convergence to the supremum. The main difference is in the direction of the inequalities: where the increasing case used  $s_N > u - \varepsilon$ , we will use  $s_N < L + \varepsilon$ . This reflects how a decreasing sequence approaches its lower bound from above.

**Proof.**

Let  $(s_n)$  be a bounded decreasing sequence. Since  $(s_n)$  is bounded, the set  $S = \{s_n : n \in \mathbb{N}\}$  is bounded below. By the Completeness Axiom, there exists an infimum  $L = \inf S$ .

Let  $\varepsilon > 0$ . Since  $L + \varepsilon$  is not a lower bound for  $S$ , there exists some  $N \in \mathbb{N}$  such that  $s_N < L + \varepsilon$ . Because the sequence is decreasing, for all  $n > N$  we have

$$s_n \leq s_N < L + \varepsilon.$$

We also know that, by definition of the infimum,  $L \leq s_n$  for all  $n$ . Combining these inequalities, we get

$$L \leq s_n < L + \varepsilon,$$

which implies that

$$|s_n - L| < \varepsilon.$$

Therefore,  $\lim s_n = L = \inf S$ .

□

**Problem 5** ( Ross Chapter 2, 10.5 )

*Prove Theorem 10.4(ii).*

**Discussion.**

Theorem 10.4 (ii) states:

*If  $(s_n)$  is an unbounded decreasing sequence, then  $\lim s_n = -\infty$ .*

To prove this, we should recall the definitions of an *unbounded* and a *decreasing* sequence.

A sequence  $(s_n)$  is called **decreasing** if for every  $n \in \mathbb{N}$ ,  $s_{n+1} \leq s_n$ . It is called **unbounded below** if for every real number  $M < 0$ , there exists an  $N \in \mathbb{N}$  such that  $s_N < M$ .

Together, these properties describe a sequence whose terms continually decrease without bound. Our goal is to show that this precisely means the limit of  $(s_n)$  diverges to negative infinity.

The strategy of the proof parallels that of part (i) of Theorem 10.4, which treats the unbounded increasing case (where  $\lim s_n = +\infty$ ). Instead of showing the sequence eventually exceeds any given positive number, we will show that for every negative number  $M$ , the sequence eventually falls below it. We will also use the monotonic property (that  $s_n$  is decreasing) to ensure that once the sequence passes below  $M$ , it stays there for all subsequent  $n$ .

**Proof.**

To show that if  $(s_n)$  is an unbounded decreasing sequence, then  $\lim s_n = -\infty$ , let  $M < 0$ .

Since the set  $\{s_n : n \in \mathbb{N}\}$  is unbounded and it is bounded above by  $s_1$ , it must be unbounded below. Hence, there exists some  $N \in \mathbb{N}$  such that  $s_N < M$ .

Because  $(s_n)$  is decreasing, for all  $n > N$  we have

$$s_n \leq s_N < M.$$

Thus, for every  $M < 0$ , there exists an  $N$  such that  $n > N$  implies  $s_n < M$ .

By definition of divergence to negative infinity, we conclude that

$$\lim s_n = -\infty.$$

□

---

**Problem 6** ( Ross Chapter 2, 10.7 )

---

Let  $S$  be a bounded nonempty subset of  $\mathbb{R}$  such that  $\sup S$  is not in  $S$ . Prove there is a sequence  $(s_n)$  of points in  $S$  such that  $\lim s_n = \sup S$ . See also Exercise 11.11.

**Discussion.** This problem asks you to construct a sequence  $(s_n)$  within  $S$  that approaches  $\sup S$ . Compare this to Exercise 11.11, which states a similar result but allows  $S$  to include its supremum and requires the sequence to be increasing. In contrast, this problem only assumes  $\sup S \notin S$ , so the challenge is to use the definition of  $\sup S$  to choose terms of the sequence that get arbitrarily close to it without ever reaching it.

**Proof.**

□